



The Pizza Optimizer

A Management Science Activity · Budget Optimization

Learning Objective

By the end of this activity, you will be able to:

- Identify the main parts of a simple management science model
- Use a spreadsheet to test solutions to a decision problem
- Define a set of choices and identify parameters
- Use binary decision variables and calculate an objective function
- Apply a budget constraint and observe how changes in inputs affect the best solution

This fits the management science idea of using quantitative analysis and mathematical models to support decision-making.

Scenario

You are in charge of ordering pizzas for a party from a trendy pizza shop. You need to choose the best possible toppings that everyone will enjoy most, but you only have **7.00ww** to spend.

Each topping has:

- **A cost** in woowoos
- **A satisfaction score**

Your job is to decide which toppings to include so that:

1. The total cost does not go over 7.00ww
2. The total satisfaction score is as high as possible

You may either choose a topping or not choose it. You cannot choose half a topping or choose the same topping twice.

This is a small example of a **management science** problem: you are trying to make the best decision under a constraint by building and testing a simple mathematical model.



Key Terms

Familiarize yourself with these terms before beginning the activity.

Management Science

A scientific, quantitative approach to decision-making that uses models to analyze a problem and find a good or optimal solution.

Model

A simplified representation of a real-world problem used to help make decisions.

Set

A collection of distinct items. In this activity, the set is the list of possible pizza toppings.

Parameter

A known value in the model. In this activity, the topping costs and satisfaction scores are parameters.

Decision Variable

A value you choose in order to make a decision. In this activity, the decision variable is whether each topping is selected or not selected.

Binary Variable

A decision variable that can only take one of two values: 1 = yes or 0 = no.

Objective Function

The formula that measures what you are trying to maximize or minimize. In this activity, the objective is to maximize total satisfaction.

Constraint

A limitation or rule the solution must satisfy. In this activity, the constraint is that the total topping cost must be \$7.00 or less.

Optimal Solution

The feasible solution that gives the best possible objective value. Here, it is the topping combination with the highest satisfaction score without going over budget.

Sensitivity Analysis

Testing how the solution changes when an input changes, such as the budget or topping data.



Standardized Data

Enter the following table into your Google Sheet:

Topping	Unit Cost (woo)	Satisfaction Score
Buffalo mozzarella	2.00ww	5.0
Gorgonzola	1.50ww	3.1
Ricotta	1.00ww	4.2
Burrata	3.00ww	4.7
Parma ham	3.00ww	7.2
Pancetta	2.00ww	4.2
Salame	2.00ww	8.3
'Nduja	0.50ww	4.8
Zucchini	1.00ww	3.5
Fried eggplant	2.00ww	5.2

Directions

Part 1 Recognize the Problem

Read the pizza scenario carefully, then answer the following:

1. Identify the decision you must make: order?
2. Identify the goal: maximize satisfaction
3. Identify the limitation: cost $\leq 7ww$
4. How will you organize this data in a spreadsheet to determine the best answer?

Part 2 Build the Spreadsheet Model

1. What is in your header?
2. What values go in each column?
3. Where does that data come from?



Part 3 Create the Formulas

1. What calculations will be needed?
2. What functions will you use?

Part 4 Test Feasible Solutions by Hand

Test possible combinations in your sheet, then post those answers here as a list of toppings:

1. Top set of toppings (26):
2. Second best set of toppings (25.6):

Part 5 Report Your Model Components

In a short written response, identify each of the following components of your model:

1. **Set:** toppings T
2. **Parameters:** price and sat Pr and Sat
3. **Decision Variables:** buy / not buy x
4. **Objective Function:** max sat.
5. **Constraint:**

Part 6 Sensitivity Analysis

1. **Price Creep:** Salame has the highest score but costs \$2. At what price does Salame stop being worth including? Increment the cost by \$0.25 until the solution changes.
2. **Satisfaction Score Tipping Point:** Pick one topping that's in the optimal (top) solution. How low can its satisfaction score drop before it gets dropped from the optimal solution? And for a topping *not* in the solution, how high would its score need to rise before it enters?



Final Reflection

Write a short response to each of the following questions:

- Was the topping with the highest satisfaction score always the best choice?
- Why is it useful to model a problem quantitatively before making a decision?
- What did you learn while completing this activity?